

From Fast Prediction to Defensible Action: A Multi-Subhypothesis Research Report on Trustworthy Transient Stability Assessment for Renewable-Rich Power Systems

Research Report Generated from the Supplied Literature Corpus
Literature synthesis and hypothesis-design report
Power-system transient stability, trustworthy machine learning, and safe control

Abstract—This report synthesizes 45 supplied papers on transient stability assessment (TSA) for renewable-energy-rich power systems. The corpus contains strong but largely parallel advances in fast data-driven assessment, uncertainty and credibility estimation, Lyapunov/barrier certificates, converter-dominated dynamics, and transfer or active learning. We formulate six complementary subhypotheses (SHs), review their evidence roles, and identify both within-SH and cross-SH gaps. The central synthesis is that present pipelines rarely close the loop from a prediction’s epistemic credibility, through a mode-dependent physical stability boundary, to an operationally feasible safety action. Two falsifiable hypotheses are proposed: (H1) map credibility into mode-conditioned physical safety margins before action; and (H2) direct the EMT simulation budget toward certificate-fragile states while a safety guard controls adaptation under drift. A staged simulation, hardware-in-the-loop, and ablation plan is provided. The proposed work deliberately distinguishes corpus coverage, compatible direct evidence, and scientific conclusion; the hypotheses are research propositions to be tested, not claims already established by the literature.

Index Terms—transient stability assessment, renewable energy, inverter-based resources, trustworthy machine learning, control barrier function, Lyapunov certificate, transfer learning, safe reinforcement learning

I. RESEARCH QUESTION AND SCOPE

Power systems with large shares of inverter-based resources (IBRs) require decisions faster than repeated high-fidelity time-domain or EMT simulation can usually provide. The supplied corpus addresses this need from several directions: supervised TSA and stability-margin prediction, probabilistic risk estimation, interpretable or credible learning, direct-method certificates, converter synchronization analysis, and adaptive learning. The broad research question is therefore:

How can a real-time TSA system turn uncertain, distribution-shifted trajectory information into a mode-aware and operationally safe decision without treating a prediction score as a physical certificate?

This report is a structured literature-derived research design rather than a systematic review claiming exhaustive external coverage. The 45 PDFs in the supplied corpus were screened by title, abstract, and selected method/limitation sections. Evidence is organized into six SHs whose functions are

intentionally different; they are not six repetitions of a single “baseline–adverse–boundary” template.

II. ITERATIVE SYNTHESIS PROCESS

The synthesis followed five iterations. First, the corpus was grouped by the scientific function of each paper rather than by a single causal chain: prediction, uncertainty/credibility, stability geometry, physical mode, control, and adaptation. Second, candidate directions that merely restated a published method were removed. Third, each remaining direction was expressed as a claim that can be evaluated by compatible evidence: simulation and calibration metrics for predictors, Lyapunov/barrier conditions for certificates, and EMT or hardware-in-the-loop (HIL) tests for converter dynamics. Fourth, interdependencies were mapped. Fifth, only gaps that have an identifiable test, comparator, and falsification condition were retained.

The resulting logic is

trajectory and operating state → prediction and credibility
→ mode-aware certificate → safe action and feedback. (1)

where every arrow has a separately testable failure mode. This avoids the common mistake of asking one article to establish prediction accuracy, physical validity, generalization, and closed-loop safety all at once.

III. MULTI-SUBHYPOTHESIS LITERATURE REVIEW AND LOCAL GAPS

Table I summarizes the six SHs. “Direct evidence” below means evidence directly compatible with the stated SH, not necessarily a controlled intervention experiment. For example, a formal Lyapunov condition is direct theoretical evidence for an invariant-set claim, whereas EMT trajectories are direct empirical evidence for a converter-transient claim.

A. SH1: Fast Assessment Is Not Yet a Decision Guarantee

The literature shows mature progress in making TSA fast and uncertainty-aware. Lu and Bu predict probabilistic post-fault rotor-angle trajectories and make their temporal distribution visible to operators [1]. Tan and Zhao address the imbalance and multimodality that make uncertainty estimates

TABLE I
SIX COMPLEMENTARY SUBHYPOTHESES, CURRENT EVIDENCE, AND USABLE GAPS

SH	Research function and literature signal	Current limitation	Usable gap and direct test
SH1	Fast TSA, margins, and probabilistic risk. Graphical learning predicts probabilistic rotor-angle trajectories [1]; debiased uncertainty targets imbalanced instability data [2]; DSPP supports chance-constrained preventive control [3].	A confidence interval or probability is usually consumed as a score, not converted into a state-dependent operational safety margin. Calibration under rare, shifted events is not equivalent to safe action.	Learn a calibrated risk envelope that explicitly penalizes false-safe decisions. Test coverage, Brier/ECE, false-safe rate, and curtailment cost under unseen faults, topologies, and renewable scenarios.
SH2	Credibility, OOD detection, and error anticipation. Hybrid credibility learning combines ensemble consistency and data-disparity indicators [4]; per-sample misclassification prediction supplies a complementary error signal [5].	Existing credibility methods can abstain or warn, but generally do not identify which physical boundary or controller constraint should become tighter.	Map credibility to a measurable uncertainty inflation and a safe-set contraction. Test whether credibility-conditioned actions dominate fixed thresholds at the same reliability target.
SH3	Physical stability certificates and safe control. Neural Lyapunov control learns a controller with a falsification module [6]; barrier-certified RL filters learned actions to meet predefined safety regions [7].	A nominal certificate can be invalidated by model mismatch, distribution shift, switching, and unmodeled converter current dynamics.	Build certificates conditioned on identified instability mode and uncertainty radius. Verify barrier/Lyapunov residuals and worst-case post-fault trajectories, not classification accuracy alone.
SH4	Hybrid and converter-dominated stability geometry. Multiple constrained stability regions model renewable trips [8]; current transients materially affect PLL synchronization assessment [9]; multi-VSC interaction changes dominant synchronization risk [10]; unstable limit cycles can form the relevant boundary [11].	A binary rotor-angle TSA label masks distinct rotor, voltage, PLL, protection/trip, and limit-cycle mechanisms. A single boundary may be physically misleading.	Infer a dominant instability mode and invoke a mode-specific certificate. Test mode accuracy and boundary-violation detection on switching, LVRT/FRT, PLL, and multi-VSC scenarios.
SH5	Transfer, active learning, and data efficiency. Bidirectional active transfer lowers the demand for labels across conditions [12]; temporal transfer addresses temporal feature mismatch [13]; physics-augmented auxiliary learning improves representation and margin calibration [14].	Label selection optimizes representativeness or classifier adaptation, not the decision-critical risk of violating a safety certificate after deployment.	Use OOD, predicted error, and certificate residual to choose EMT labels. Test labels-to-safety efficiency against random, entropy, and OOD-only sampling.
SH6	Operational integration. Chance-constrained preventive control provides a link between uncertainty and set-points [3]; safe RL offers a control filter [7].	Assessment, certification, control, and post-event model update remain modular; no clear contract determines when an uncertain forecast must trigger conservative control, further simulation, or abstention.	Define a three-action policy: act, constrain, or request high-fidelity evaluation. Test closed-loop security, recovery, and operational cost.

unreliable exactly where rare instability matters [2]. Su *et al.* place a Bayesian surrogate inside chance-constrained preventive control [3]. These are essential ingredients, but they leave a decision gap: the threshold converting a statistical output into an action is often fixed, although prediction error and operating consequences are state-dependent.

The SH1 gap is therefore not “improve accuracy” in the abstract. It is to estimate a conservative decision risk $\mathcal{R}(x)$ whose calibration is evaluated especially on the false-safe tail. A negative result is informative: if a calibrated risk envelope cannot outperform a single threshold at matched curtailment cost, the additional machinery does not justify operational use.

B. SH2: Credibility Must Be Actionable

Credibility research correctly recognizes that high indistribution accuracy does not establish reliability under changing conditions. Hybrid credibility learning uses ensemble behavior, outlier degree, and one-sample MMD, including misclassified OOD examples in training [4]. Misclassification prediction supplies a per-sample probability that an otherwise fast TSA classifier is wrong [5]. However, a warning has no intrinsic operational meaning. The system must decide

whether to keep the nominal set-point, impose a conservative constraint, or obtain more authoritative evidence.

The SH2 gap is an explicit calibration-to-action contract. Let $c(x) \in [0, 1]$ be credibility and $u(x)$ a calibrated uncertainty radius. The report proposes that $u(x)$ changes the admissible safety set rather than merely coloring a dashboard:

$$\mathcal{S}_{\text{rob}}(x) = \{a : \max_{\delta \in \Delta(u(x))} B_{m(x)}(f(x, a, \delta)) \geq 0\}, \quad (2)$$

where $B_{m(x)}$ is a mode-specific barrier function and Δ expands as credibility decreases. This equation is a testable design proposal, not a claim that the available credibility models already ensure safety.

C. SH3: Certificates Are Strong but Need Deployment Validity

Lyapunov and barrier methods contribute a qualitatively different kind of evidence from a predictor: under their stated assumptions, they establish invariance or decrease conditions. Neural Lyapunov control jointly searches for a control law and a candidate Lyapunov function, with falsification used to enforce conditions [6]. Barrier-certified RL similarly filters learned actions against a neural barrier certificate [7]. Their limitation in the combined setting is external validity: a

certificate for a nominal model does not necessarily remain valid when input trajectories are shifted or when the relevant physical mode has changed.

The local gap is a certificate that declares its domain of validity. It should report the physical mode, uncertainty radius, residual violation statistics, and whether it was verified by RMS simulation, EMT simulation, or HIL. This separates theoretical validity from empirical support rather than combining them into one opaque score.

D. SH4: The Stability Boundary Is Hybrid and Mode-Dependent

The physical-literature branch warns against treating all insecurity as the same rotor-angle classification problem. Mishra *et al.* model prone-to-trip renewable-rich systems as switched systems with multiple LVRT-constrained stability regions [8]. Chen *et al.* show that neglecting converter current transients can cause inaccurate synchronization assessment [9]. Wang *et al.* further show that nonlinear damping and transient interaction among VSCs determine dominant-converter risk [10]. Xu *et al.* explain that an unstable limit cycle, not only an unstable equilibrium point, may shape the relevant VSC boundary [11].

The usable SH4 gap is a *hybrid boundary atlas*: a mode classifier maps a trajectory to rotor-angle, voltage, PLL-synchronization, protection/trip, or limit-cycle-dominated regimes, and each regime uses an appropriate stability function. A rigorous failure case would be mode confusion that increases the false-safe rate relative to a single robust classifier; this must be measured, not assumed away.

E. SH5: Adaptation Should Spend Labels Where Safety Is Most Fragile

Transfer and active learning reduce the cost of adapting to operating changes. Bidirectional active transfer uses both useful target instances and selective removal of source instances [12]; temporal transfer specifically aligns evolving time-series features, including zero-target-sample use cases [13]. Physics-augmented auxiliary learning offers a mechanism-guided route to margin prediction and generalization [14]. Yet a label selected because it is uncertain or representative may be operationally unimportant, while a rare trajectory close to a certificate violation can be decisive.

The SH5 gap is a safety-value acquisition function. Instead of selecting only by entropy, choose the event whose high-fidelity label is expected to most reduce robust-boundary violation. The key result is not average accuracy at a fixed label budget, but the area under a labels-versus-false-safe curve.

F. SH6: An Operational Contract Is Still Missing

The existing pieces suggest, but do not complete, an online operating loop. A credible predictor may flag OOD; a physical certificate may filter an action; a preventive optimizer may adjust set-points. The missing contract is what happens when these disagree. In particular, a low-confidence but nominally stable prediction should not automatically be labeled safe,

and a conservative certificate need not force unnecessary curtailment if a rapid EMT/HIL check is available.

The SH6 gap is a policy with three outcomes: *act* when risk is low and certification holds; *constrain* when risk is elevated but a robust action exists; and *escalate* when credibility or certificate residual is inadequate. This turns model uncertainty into an auditable engineering workflow.

IV. DETAILED GAPS FOR CONVERTER-DOMINATED, RENEWABLE-RICH GRIDS

The following refinement makes the gap statements operational for high-renewable-penetration grids with power-electronic equipment. The quantitative values in this section are *reported evidence anchors* from the supplied papers, not numbers that are assumed to transfer unchanged across networks, fault-clearing practices, or controller settings. Cross-paper accuracy is not treated as a leaderboard because the fault sets, class balance, PMU windows, and security labels differ.

A. SH1: Risk and Margin Must Retain Physical Units

What is established. Fast predictors are now sufficiently capable that their output can be used inside an online loop. A hierarchical PMU-based model reported an average response time of 16 ms over 3000 test cases, a stability-margin-prediction error below 0.036, and accuracy above 98.5% on its reported unknown scenarios [20]. An improved deep Lyapunov learner reported 98.95% overall accuracy and 99.28% precision on its test case [15]. For probabilistic assessment at scale, PPGP was evaluated on a Texas 2000-bus system with 471 PV and 470 wind generations [16]; a separate large-scale EMT framework reported approximately 400-fold acceleration for a synthetic IEEE 118-bus AC/DC system containing two million RES entities [17]. These results establish computational feasibility, not a universal safety guarantee.

Precise gap. The common output is a class probability, a transient stability index, a critical clearing time (CCT), or a scalar margin. Such values are often not tied to the *physical limiting event*: first-swing angle separation, voltage recovery failure, PLL loss of synchronism, current saturation, LVRT/FRT tripping, or a controller state reaching its hard limit. Consequently, a well-calibrated probability can still be false-safe if the latent model does not represent the active boundary.

Researchable SH1 claim. For each contingency e , estimate a vector

$$\mathbf{y}_e = [\widehat{\text{CCT}}, \widehat{\Delta\delta}_{\max}, \widehat{\Delta V}_{\min}, \widehat{\omega}_{\text{PLL}}, \widehat{I}_{\max}, \widehat{p}_{\text{trip}}], \quad (3)$$

along with prediction intervals. The margin used for action is the minimum normalized distance to the applicable protection, synchronization, voltage, and thermal/current constraints, rather than an unqualified TSA label. CCT estimation work already treats the CCT as a stability-boundary quantity [18]; trajectory-sensitivity work shows that sensitivity norms can rank control and reference parameters that move the DFIG-WTG synchronization boundary [19].

Quantitative test and falsifier. Stratify test events by CCT distance, IBR penetration, short-circuit ratio, fault duration/location, and control mode. Report root-mean-square error for every physical component in (3), 90% interval coverage, expected calibration error (ECE), and false-safe rate within the most critical 10% of the CCT/margin distribution. A planned success criterion is a 95% bootstrap upper confidence bound on false-safe rate below 1% in the critical stratum; failure of coverage or a false-safe concentration near a limiter invalidates the proposed margin mapping even if global accuracy remains high.

B. SH2: Credibility Requires a Decomposition, Not One Threshold

What is established. Credibility can be learned from model consistency and distributional disparity [4]; scene-dependent credibility has also been formulated using a localized generalization-error bound and a stability-density construction. In its reported cases, a critical scene-credibility index of 0.93 separated results with 100% stated accuracy [21]. These methods are valuable alarms, but their scalar score does not distinguish sensor corruption, operating-point extrapolation, topology change, label ambiguity near a boundary, or absent converter physics.

Precise gap. A single credibility score can conceal materially different operational responses. PMU noise or a missing feature should request data-quality checking; a novel grid strength or PLL parameter should enlarge model uncertainty; an imminent hard-limit encounter should immediately shrink the feasible control set. In addition, a credibility classifier trained on rotor-angle labels may look confident while the actual event is short-term voltage instability or converter synchronization loss.

Researchable SH2 claim. Replace $c(x)$ by

$$c(x) = [c_{\text{data}}, c_{\text{shift}}, c_{\text{mode}}, c_{\text{model}}, c_{\text{certificate}}], \quad (4)$$

where the entries respectively measure data integrity, OOD distance, uncertainty in dominant-instability-mode (DIM) assignment, predictive disagreement, and certificate residual. The policy must state which component triggered escalation. This is supported by evidence that a networked shapelet TSA retained over 98% reported accuracy under studied cases and lost less than 0.7 percentage points under network-parameter errors, while its PMU-noise tests explicitly ranged from 0.5% to 3% [22]; that robustness observation should not be extrapolated to unmodelled IBR control changes.

Quantitative test and falsifier. Construct separated stress suites: PMU noise/dropout, line/topology change, grid-strength change, renewable forecast error, and PLL/current-control mismatch. Compare one scalar threshold with the vector policy in (4) at equal intervention cost. Report per-source AUROC for error detection, ECE, mode-conditional coverage, and the percentage of escalations that are later verified necessary by EMT. The decomposition fails if it raises the total escalation rate without improving calibration, false-safe rate, or attribution correctness.

C. SH3: Certificate Validity Must Include Saturation and Converter Timescales

What is established. Neural Lyapunov and barrier methods provide a formal safety mechanism under their stated state-space and model assumptions [6], [7]. Deep Lyapunov learning further links the learned decision function to a region-of-attraction construction [15]. Yet a power-electronic grid is a differential-algebraic, switched and constrained system: current controllers, PLLs, DC-link/AC voltage dynamics, virtual inertia, ride-through logic, protection, and anti-windup limiters introduce mode switches and time-scale separation.

Precise gap. A certificate $V(x)$ or $B(x)$ learned from nominal RMS trajectories may remain positive while the real control action enters current limitation or a PLL transient invalidates the reduced model. The hard-limit formulation of Romay *et al.* shows that control bounds must be enforced inside the dynamic equations through complementarity conditions, rather than checked after a trajectory is generated [23]. This exposes a missing contract between learned certificates and constraint-active dynamics.

Researchable SH3 claim. Use a hybrid certificate bank $\{B_m, V_m\}$ indexed by both DIM and limiter state m :

$$\dot{V}_m(x, a) \leq -\alpha_m \|x - x_m^*\|^2 + \rho_m u(x), \quad B_m(x) \geq 0, \quad (5)$$

where $u(x)$ is the credibility-derived uncertainty radius. Mode transitions must satisfy reset or dwell-time conditions, and each certificate must declare whether its evidence comes from analytical residual checks, RMS simulation, EMT simulation, or HIL.

Quantitative test and falsifier. Record the maximum violation of (5), time spent in current/voltage saturation, PLL frequency/angle excursion, and violation rate after a mode switch. Compare a nominal smooth certificate, a hard-limit-aware certificate, and the hybrid bank. The SH is not supported if robust filtering avoids violations only by excessive load shedding/curtailment, or if certificate residuals cease to predict EMT violations under current transients.

D. SH4: Dominant Instability Modes Need a Boundary Atlas

What is established. High-renewable systems can exhibit coupled but nonidentical rotor-angle, short-term voltage, and converter-synchronization phenomena. DIM work explicitly targets the distinction between rotor-angle and short-term voltage instability. A local-global GNN/Transformer formulation found its peak reported accuracy with a 2 s window and training time below 3 s per epoch, and illustrated that DIM-guided control selects generator shedding versus load shedding differently [24]. The multi-VSC study includes non-linear damping, angular-frequency jumps, and transient interactions, with PSCAD/EMTDC and RT-LAB validation [10]. PLL current transients cannot be neglected without distorting synchronization assessment [9]. Finally, an unstable limit cycle can be the relevant VSC stability boundary rather than an unstable equilibrium point [11].

Precise gap. Current DIM methods mostly assign a label, while direct methods mostly analyze one specified mechanism.

The missing object is a *boundary atlas* that maps a trajectory to (i) a dominant mechanism, (ii) the correct reduced state and observable, (iii) a validity domain, and (iv) an appropriate remedial action. Without it, a rotor-angle surrogate may be asked to judge an LVRT trip cascade, and a voltage-oriented action may be applied to a PLL-dominated event.

Researchable SH4 claim. Let $m \in \{\text{RA, STV, PLL, trip, ULC}\}$ denote rotor-angle, short-term voltage, PLL synchronization, protection/trip, and unstable-limit-cycle mechanisms. Learn $p(m | z)$ from PMU and converter features, but use mode-specific observables and action sets: rotor-angle separation/CCT; voltage nadir and recovery time; PLL angle-frequency/current excursion; LVRT/FRT dwell/trip count; and center-manifold/limit-cycle distance, respectively. The confidence of this assignment becomes the c_{mode} term in (4).

Quantitative test and falsifier. Evaluate macro-F1 and a cost-weighted confusion matrix, not only overall DIM accuracy. Every misclassification must be labeled as benign, conservative, or false-safe after EMT replay. Required scenario axes include IBR share, short-circuit ratio, number of interacting VSCs, fault impedance/duration, PLL bandwidth, current limit, LVRT/FRT setting, and protection delay. A conservative planning target is to reduce false-safe cross-mode errors by at least 30% relative to a binary TSA baseline at matched remedial-action energy; the atlas is rejected if extra modes merely redistribute errors without lowering the false-safe cross-mode subset.

E. SH5: Data Adaptation Should Target Certificate Failure, Not Only Classifier Uncertainty

What is established. Multiple techniques reduce simulator and labeling cost. Transfer ELM reported more than 24% improvement in stability-assessment accuracy in its large-scale cross-scale transfer setting [27]. Instability-pattern-guided updating began from 98.37% test accuracy with 181 errors and used a 556-sample representative set, about 5% of its original dataset, to constrain update cost [25]. A topology-clustering/DCGAN study reported, for one scenario, accuracy/security/recall of 97.40/97.81/88.10%, compared with 91.64/92.14/80.13% for random oversampling, but explicitly notes that its target is global transient power-angle stability rather than other instability modes [26]. These results show that sample efficiency is possible, while also revealing the class and mode scope of current targets.

Precise gap. Entropy, margin, or OOD sampling may spend an EMT label on a novel but operationally harmless trajectory, while undersampling a familiar-looking event near a hard current limit or mode transition. Updating a classifier on a pattern can also cause certificate drift or catastrophic forgetting in regions that were previously safe.

Researchable SH5 claim. Schedule high-fidelity labels with

$$A(z) = \alpha d_{\text{OOD}}(z) + \beta \hat{p}_{\text{err}}(z) + \gamma r_{\text{cert}}(z) + \eta H[p(m | z)] + \kappa L_{\text{impact}}(z), \quad (6)$$

where r_{cert} is a barrier/Lyapunov residual and L_{impact} estimates load loss, curtailment, or protection consequence. The update has a replay/constraint term that preserves previously verified certificate margins.

Quantitative test and falsifier. Build learning curves of EMT labels versus false-safe rate, mode-macro-F1, ECE, certificate violations, and retained performance on prior scenarios. Compare random, entropy, margin, OOD-only, and (6) under identical EMT budgets. A pre-registered target is at least 25% fewer new EMT labels to reach the baseline's false-safe rate, with no greater than one percentage-point loss on the source-regime safety score. Failure on either the safety or forgetting condition rejects the active-transfer claim.

F. SH6: Control Must Optimize Security, Recovery, and Feasibility Together

What is established. Control-oriented TSA is no longer only conceptual. A master-slave TSC-OPF framework reports millisecond real-time dispatch on New England and Australian systems while incorporating contingency-specific surrogate feedback [28]. An integrated TSA-monitoring approach uses spatial-temporal features and critical-generator identification to select generator tripping for impending instability [29]. For converter control, a multi-VSG plus resistive superconducting fault-current-limiter study reported 62.12% current limiting during fault, 41.24% after clearing, a maximum 3.3% power vacancy, and 44.59% less power vacancy than its DDPG comparator in the reported cases [30]. These are valuable proof points, but they optimize different operational objectives under different mechanisms.

Precise gap. Existing control integrations generally assume a correct TSA/DIM result and do not expose what the controller should do when credibility is low, the mode is ambiguous, or a certificate is invalid. They also rarely compare generator tripping, load shedding, reactive support, VSG virtual-inertia/damping set-points, current-limit management, and topology actions on a common false-safe/cost scale.

Researchable SH6 claim. Solve a receding-horizon, three-action decision:

$$\min_{a \in \mathcal{A}_m} J(a) = w_s \hat{p}_{\text{FS}}(a) + w_e E_{\text{shed}}(a) + w_r T_{\text{recovery}}(a) + w_c C_{\text{wear}}(a), \quad (7)$$

subject to the appropriate hybrid certificate, equipment limits, and a latency budget. If no feasible robust action exists, escalate to EMT/HIL or a predefined protection policy. The objectives make a false-safe declaration more costly than a conservative action, while preventing a nominally safe controller from achieving its score through indiscriminate curtailment.

Quantitative test and falsifier. Report delivered load/energy, renewable curtailment, generator/VSC trips, peak fault current, voltage recovery time, PLL resynchronization time, CCT improvement, and end-to-end decision latency. Compare with fixed rules, TSA-only remedial action, chance-constrained OPF, and nominal barrier RL. The control integration is unsuccessful if it cannot satisfy the selected

protection/ride-through constraints within the latency budget, or if equal-security comparisons show no benefit beyond additional shedding.

V. DEPLOYMENT AND STRESS-TEST CONTRACTS

A. Latency Contract: A Cascaded Decision, Not a Simultaneous Stack

The reported 16 ms response of a particular TSA model [20] is not evidence that an encoder, probabilistic predictor, five-part credibility vector, DIM classifier, certificate bank, robust filter, and active-learning score can all run serially within 16 ms. This report therefore does *not* claim a universal 16 ms end-to-end deadline. Instead, each target installation declares a protection or dispatch deadline T_{deadline} , and the proposed controller uses an early-exit cascade:

- 1) a shared, fixed-cost feature and data-integrity stage produces a low-cost risk bound;
- 2) low-risk, in-distribution events use a pre-verified nominal safe action;
- 3) only boundary-near, OOD, or high-consequence events invoke DIM selection and the mode-specific certificate;
- 4) certificate repair, EMT label acquisition, and meta-parameter updating are asynchronous and never sit on the protection-critical path.

The hard real-time contract is

$$T_{\text{feature}} + T_{\text{base}} + \mathbb{I}_{\text{escalate}}(T_{\text{DIM}} + T_{\text{cert}} + T_{\text{filter}}) \leq T_{\text{deadline}}, \quad (8)$$

where the indicator is determined before the expensive branch. The measured evidence must include p50, p95, p99, and maximum latency, memory/compute utilization, and decision-quality metrics for every branch. A claimed 16 ms configuration is falsified if its p99 or its worst observed safety-critical response exceeds 16 ms; reducing simulation fidelity, sampling rate, or certificate residual checks to pass timing is not an acceptable remedy unless the resulting physical coverage is separately re-established.

B. Physical Calibration of the Credibility-to-Robustness Map

The uncertainty radius must not be an uncalibrated rule such as $u = 1/c$. For mechanism m and physical constraint j , define the signed physical residual $r_{m,j}(x)$ in its native engineering unit: e.g., CCT reserve in ms, voltage reserve in p.u., PLL angle/frequency reserve, current-limit reserve in p.u., or ride-through dwell-time reserve. A mode-conditional calibrated lower bound is

$$\begin{aligned} \underline{r}_{m,j}(x) &= \hat{r}_{m,j}(x) - q_{m,j}(\phi(x)), \\ \Pr\{r_{m,j}(x) \geq \underline{r}_{m,j}(x) \mid m, \phi\} &\geq 1 - \epsilon_{m,j}, \end{aligned} \quad (9)$$

where $q_{m,j}$ is a split-conformal or other finite-sample calibrated quantile and ϕ contains operating state, IBR penetration, grid strength, observed communication quality, and OOD score. The robust controller enforces $\underline{r}_{m,j}(x, a) \geq 0$ for all active constraints. Thus the role previously denoted by $u(x)$ is a vector of empirically validated, constraint-specific physical

buffers, rather than a Bayesian neural-network variance or an arbitrary scalar inflation.

“Separate” calibration must not mean a sparsely populated quantile for every raw combination of DIM, constraint, topology, and bus. In a multi-VSC system, that Cartesian product rapidly becomes untestable. The design therefore declares a finite calibration cell $g = (m, h(j), c)$ before evaluation: $h(j)$ is a physically justified constraint family (for example, PLL/current/LVRT rather than an arbitrary bus identifier), and c is a declared critical-interface, operating-regime, and data-quality cluster. Every reported cell must give its EMT sample count n_g , held-out coverage, and sharpness. A cell with $n_g < n_{\min}$ receives neither a finely tuned nor an optimistic local quantile: it must use a predeclared conservative family-level envelope or escalate. This preserves a falsifiable marginal-coverage claim rather than silently pretending that dozens of sparse calibrators have conditional coverage.

If $\phi(x)$ lies outside the calibration support, the method is not permitted to extrapolate an optimistic buffer: it must issue a conservative pre-verified fallback or escalate. The validation report must give coverage and sharpness for each declared cell, coverage near the CCT/current/voltage boundary, and the rate at which an OOD event is incorrectly accepted as calibrated. The full lifecycle label ledger remains transparent:

$$\begin{aligned} N_{\text{EMT}}^{\text{total}} &= N_{\text{base}}^0 + \Delta N_{\text{adapt}}, \\ N_{\text{base}}^0 &= N_{\text{cal}}^0 + N_{\text{safety}}^0, \\ \Delta N_{\text{adapt}} &= \Delta N_{\text{acq}} + \Delta N_{\text{cal}} + \Delta N_{\text{safety}}, \end{aligned} \quad (10)$$

where N_{cal}^0 and N_{safety}^0 are the one-time, pre-deployment calibration and independent-safety reserve, shared by every arm; ΔN_{acq} is the post-deployment label investment chosen by H2; and ΔN_{cal} , ΔN_{safety} expose any additional recalibration or safety-monitoring burden caused by drift. The preregistered primary H2 endpoint is marginal label efficiency: the false-safe learning curve against ΔN_{acq} , with a common N_{base}^0 and equal declared coverage target. The fixed baseline and the complete ledger in (10) are reported separately as lifecycle economics. Thus an extra 5000 EMT trajectories needed merely to stabilize calibration is neither silently free nor incorrectly charged to an H2 selector that acts only on rare post-deployment boundaries.

C. Drift-Resilient H2: Constrained Online Meta-Adaptation

The H2 weights cannot remain fixed after offline tuning because topology, renewable dispatch, grid strength, and IBR-control settings drift. Let $\theta_t = (\alpha_t, \beta_t, \gamma_t, \eta_t, \kappa_t)$ be the acquisition parameters for OOD novelty, predicted error, certificate residual, mode uncertainty, and consequence. An online meta-learner proposes θ_{t+1} only after a delayed batch of authoritative EMT labels; its context includes recent calibration residuals, mechanism mix, shift statistics, and action outcomes. The fast path uses frozen θ_t , whereas the update path is asynchronous.

The update is accepted only if an independent rolling safety set satisfies a high-confidence constraint, for example

$$\text{UCB}_{1-\delta}(\text{FSR}_{t+1}) \leq \text{UCB}_{1-\delta}(\text{FSR}_t) + \xi, \quad (11)$$

and if calibrated coverage and source-regime retention stay above their declared floors. To prevent noisy rare-event estimates from creating a rollback/update oscillation, the experiment predeclares a minimum update interval $\Delta_{\min}$ and a nonzero statistical deadband d_{DB} . Let t_ℓ be the last accepted update, let “ref” denote the frozen reference window, and define $F_t^L = \text{LCB}_{1-\delta}(\text{FSR}_t)$ and $F_{\text{ref}}^U = \text{UCB}_{1-\delta}(\text{FSR}_{\text{ref}})$. A proposal is even evaluated only when

$$\text{UpdateEligible}_t = 1 \iff \begin{aligned} &t - t_\ell \geq \Delta_{\min}, \\ &F_t^L > F_{\text{ref}}^U + d_{\text{DB}}. \end{aligned} \quad (12)$$

Otherwise the approved θ_t is retained and the system accumulates evidence; no EMT-driven weight churn is credited as adaptation. Every evaluation must report $\Delta_{\min}$, d_{DB} , proposal, acceptance, and rollback counts, and the sensitivity of safety to both values. The evaluation compares fixed offline coefficients, unconstrained online meta-adaptation, and the gated constrained update in (11)–(12) under controlled nonstationary shifts. This converts online meta-learning from a performance slogan into a falsifiable safety-preserving adaptation mechanism.

D. Cost Comparison as a Pareto Frontier

Fixed weights in (7) can obscure an unsafe or excessively conservative policy. Rather than declaring success at one selected (w_s, w_e, w_r, w_c) , estimate the attainable frontier over the vector

$$\mathbf{g}(a) = \begin{bmatrix} \text{FSR}^{\text{UCB}}, E_{\text{unserved}}, E_{\text{curtail}}, \\ T_{\text{recovery}}, N_{\text{trip}}, C_{\text{wear}} \end{bmatrix}^T. \quad (13)$$

Use epsilon-constraint sweeps: minimize operational disruption subject to a predeclared false-safe upper bound, protection/ride-through constraints, and a latency limit; then repeat across safety bounds. Compare the resulting Pareto front to each baseline through non-dominance rate, hypervolume, and paired scenario differences. H1 is supported only if it expands the feasible safe region or Pareto-dominates a baseline for a material scenario subset. Eliminating false-safe events solely through disproportionate shedding appears as a different point on the frontier, not as an automatic win.

E. Training, Independent Falsification, and Hidden Communication Stress

The evidence hierarchy is explicit. Broad RMS simulation explores operating space and proposal ablations. EMT simulation produces scalable, high-fidelity pseudo-labels for training and calibration, especially near boundaries and under converter current/PLL dynamics. RT-LAB/HIL is *not* the source of millions of training labels: it is a frozen, independent set of sparse but deliberately difficult falsification points for checking model, timing, interface, and control-implementation mismatch.

The held-out stress protocol includes ordinary noise and topology changes, plus communication and actuation faults that are realistic in PMU/control networks: time-varying latency and jitter, correlated measurement dropout, burst packet loss, time-stamp skew, delayed or missing topology updates, replay-like stale measurements, and loss of one or two control-command cycles. The latter must be parameterized at the actual system frequency, so one and two cycles correspond to 20/40 ms at 50 Hz or 16.7/33.3 ms at 60 Hz. A lost command is not treated as a nonexistent action: the actuator must execute a specified hold-last policy a_{hold} , with the last valid command applied over the loss interval. The certificate is valid only if

$$\min_{0 \leq \tau \leq \tau_{\text{hold}}} r_{m,j}(x_\tau; a_{\text{hold}}) \geq 0 \quad \text{for all active } (m,j), \quad (14)$$

where τ_{hold} is the declared worst case (40 ms for the two-cycle 50 Hz case, and no smaller than the actual tested delay). Tests must include correlated failures rather than independent missing samples only. Each hidden case is scored for detection delay, erroneous acceptance, fallback correctness, peak current/voltage/PLL excursion, and whether the command reaches an active limiter after it is too late. This prevents a model from appearing reliable merely because its train/test split lacks the failure modes operators face.

F. Scalability Contract: Critical Interfaces, Not a Global Certificate

The Texas 2000-bus PPGP result [16] establishes that a large-scale probabilistic TSA calculation can be evaluated; it does not establish that a global DIM classifier and a mode-constraint certificate bank scale linearly over every bus and interface. The proposed deployment contract therefore does *not* construct a 2000-bus tangent-plane or residual model at every location. Offline fault and sensitivity screening identifies a predeclared set of critical interfaces: weak-grid VSC point-of-common-coupling neighborhoods, major transfer corridors, voltage pockets, and protection boundaries. The escalated branch evaluates only the interface templates activated by the shared low-cost screen, using local dynamic states plus a declared external-network equivalent.

For each network size, the report must disclose the number of monitored interfaces, DIM classes, active constraint families, calibration cells, and maximum concurrently escalated templates. It must then report the nominal and escalated p99/max latency and coverage by interface. A large-system scalability claim is falsified if the critical-interface branch cannot meet its declared deadline, if a topology change activates an unmodeled interface without conservative escalation, or if coverage is reported only after dropping difficult interfaces. This makes the dimensionality reduction an auditable modeling assumption rather than an unreported source of speed.

G. Irreducible-Pessimism Audit

An extreme protective action is sometimes unavoidable, but it must not disappear into an aggregate false-safe statistic. Define an irreducible-pessimism (IP) event when every available mode score is low-confidence, $\max_m c_{\text{mode},m}(x) < c_{\min}$,

and the calibration context lies outside every declared support, $\phi(x) \notin \bigcup_g \Phi_g$. The policy must then take a predeclared all-system safe action a_{IP} (for example, emergency curtailment or a protective trip), rather than label the event as confidently safe. Its held-out incidence is

$$\rho_{\text{IP}} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \mathbb{I}_{\text{IP}}(x_i). \quad (15)$$

The report must separately give IP count, ρ_{IP} , unserved energy/curtailment, trips, and recovered physical margins; these events cannot be counted as ordinary successful abstentions. Before testing, set $\rho_{\text{IP},\text{max}} = 5\%$ for the held-out suite. If $\rho_{\text{IP}} > 5\%$, the critical-interface screen is deemed to have omitted too much unknown behavior: its scope and OOD/escalation thresholds must be reopened before any deployability claim is made.

VI. EXISTING RESEARCH LIMITATIONS AT THE COLLECTION LEVEL

The six SHs expose four collection-level limitations.

- 1) **False-safe asymmetry is underrepresented.** Average accuracy, even with standard uncertainty intervals, does not reflect the much higher cost of incorrectly declaring a post-fault state safe.
- 2) **Epistemic and physical uncertainty are decoupled.** OOD or misclassification indicators identify when a predictor may be unreliable, but are not generally propagated into a physical safe set.
- 3) **Mechanism heterogeneity is flattened.** Converter synchronization, rotor-angle separation, ride-through trip, and limit-cycle boundaries are often evaluated with distinct tools but merged into one TSA output.
- 4) **Adaptation is not certificate-aware.** Data-efficient transfer selects informative data, but normally does not prioritize the trajectories that make the deployed controller or certificate least defensible.

These are not criticisms of individual papers. They arise because the papers solve different necessary subproblems. The research opportunity lies in integrating them while preserving each method's evidentiary limits.

VII. COMBINATION GAP AND GENERATED HYPOTHESES

A. Combination Gap

No direct synthesis in the supplied corpus simultaneously links (i) per-event credibility and false-safe risk, (ii) a mode-dependent hybrid stability boundary, (iii) a certificate-filtered operational action, and (iv) label acquisition driven by future certificate risk. Pairwise combinations exist: uncertainty with preventive control [3], credibility with OOD assessment [4], and certificate-based control [7]. The four-way linkage is the combination gap. It is scientifically useful because a downstream action can be safe only if the uncertainty, physical boundary, and deployment update refer to the same operating state and uncertainty assumptions.

B. Hypothesis H1: Credibility-Calibrated Hybrid Stability Certificate

H1. A cascaded TSA policy that maps credibility into mode-conditioned physical safety margins before certificate-filtered action will improve the empirical safety-disruption Pareto frontier over nominal certificates, pure TSA thresholds, and credibility-only abstention under topology, fault, IBR-control, and communication shifts.

The causal proposal is: PMU/trajectory features and operating context $\rightarrow$ calibrated physical residual bounds plus credibility decomposition $\rightarrow$ dominant instability mode m and constraint-specific robust buffers $\rightarrow$ certificate-filtered control set $\mathcal{S}_{\text{rob}}$ $\rightarrow$ realized post-fault security. Under command loss, that control set must also satisfy the hold-last tolerance in (14); a certificate that assumes an undelivered command is not an H1 success. Cost comparison uses the frontier in (13), not one matched scalar cost: a method should not appear safer merely by curtailing all controllable resources.

H1 has clear counterevidence. It is rejected or narrowed if it fails the latency contract in (8), if calibrated residual coverage collapses under unseen mode/communication shifts, if the hold-last condition in (14) violates a physical limit, if ρ_{IP} exceeds the preregistered 5% limit in (15), if its Pareto front is dominated by a baseline, or if the mode selector causes more false-safe boundary errors than the nominal safe policy.

C. Hypothesis H2: Certificate-Violation-Risk-Driven Active Transfer

H2. Under operating drift, a safety-guarded online meta-learner that directs marginal EMT label investment to certificate-fragile states will reach a target false-safe rate with fewer post-deployment selected labels than fixed-weight random, entropy, margin, or OOD-only active transfer, without degrading a rolling safety hold-out set.

For candidate event z , the fast path uses the currently approved coefficient vector θ_t in the acquisition score

$$\begin{aligned} A_{\theta_t}(z) = & \alpha_t d_{\text{OOD}}(z) + \beta_t p_{\text{err}}(z) \\ & + \gamma_t r_{\text{cert}}(z) + \eta_t q_{\text{mode}}(z) \\ & + \kappa_t L_{\text{impact}}(z), \end{aligned} \quad (16)$$

where d_{OOD} denotes distributional novelty, p_{err} predicted classification error, r_{cert} a barrier/Lyapunov residual or near-violation score, q_{mode} rewards underrepresented mechanism modes, and L_{impact} captures consequence. Coefficients are updated asynchronously only after EMT labels arrive, only after the update gate in (12) opens, and only if the acceptance constraint in (11) passes; the online decision path never waits for meta-learning.

H2 is falsified if its lower ΔN_{acq} is achieved only through more conservative operation, if its constrained update rejects most drifted regimes, if it forgets previously reliable regimes, if it requires materially larger ΔN_{cal} or ΔN_{safety} than a matched baseline, or if its safety rate is no better than simpler fixed-weight acquisition baselines at equivalent marginal selected-label budget.

VIII. FEASIBLE EXPERIMENTAL PROGRAMME AND EXPECTED RESULTS

A. Testbed and Data Protocol

Use three escalating testbeds: IEEE 39-bus and IEEE 118-bus systems for comparability with current PTSA studies [1]; a renewable-rich switched-system case with LVRT/FRT trip logic [8]; and a multi-VSC network with PLL, current-transient, nonlinear-damping, and interaction effects [9], [10]. RMS simulation explores broad operating space. EMT generates scalable high-fidelity pseudo-labels for training, calibration, and active-learning updates, especially near CCT, current-limit, voltage-recovery, and PLL boundaries. A frozen set of RT-LAB/HIL cases is reserved for sparse but independent falsification of timing, interfaces, model mismatch, and control implementation; it is not used as a bulk label generator.

Partition data by *scenario*, not random trajectory: hold out fault locations, clearing times, renewable penetration ranges, line outages, short-circuit ratio, PLL parameters, current limits, control modes, and topologies. This prevents the misleading result in which nearly identical trajectories appear in both training and testing. Record a mechanism label from simulation diagnostics: rotor separation, voltage violation, PLL loss of synchronization, protection/trip cascade, or limit-cycle proximity. Create a final hidden stress suite with PMU jitter, correlated channel loss, burst packet drops, clock skew, stale topology information, and delayed or missing control commands. Command-loss tests explicitly cover one and two cycles at the operating frequency, namely 20/40 ms at 50 Hz and 16.7/33.3 ms at 60 Hz, and propagate the actual hold-last actuator trajectory through the physical constraints in (14); they do not assume that a lost command was executed.

B. Models, Baselines, and Ablations

The full method is a shared-feature, early-exit cascade rather than a monolithic serial stack: a fixed-cost data-quality and base-risk branch serves all events; DIM/certificate/filter branches run only after escalation; and active-transfer/meta-learning operates asynchronously. In the Texas 2000-bus case, the escalated branch is restricted to predeclared critical-interface templates rather than a global certificate evaluation. Compare against: (B1) deterministic TSA threshold; (B2) uncertainty-aware TSA without physical-residual calibration; (B3) credibility abstention without a certificate; (B4) nominal neural Lyapunov/barrier control; (B5) fixed-weight active transfer using random, entropy, margin, and OOD-only acquisition; (B6) a monolithic all-modules serial pipeline, which tests whether apparent safety gains are achievable within the declared latency budget; and (B7) an unpartitioned global certificate/DIM implementation, which tests whether the critical-interface reduction is necessary for scale.

Remove one component at a time: physical residual calibration, credibility decomposition, mode-specificity, hard-limit awareness, robust filtering, certificate-residual acquisition, the $\Delta_{\min}/d_{\text{DB}}$ update gate, asynchronous meta-adaptation, critical-interface reduction, and physics auxiliary loss. The

TABLE II
PRIMARY OUTCOMES AND SUCCESS CRITERIA

Outcome	Interpretation
False-safe rate	Fraction of declared-safe events that violate the authoritative EMT/HIL security criterion; primary safety outcome.
Risk calibration	ECE, Brier score, and risk-coverage curves, reported overall and near the stability boundary.
Certificate validity	Frequency/magnitude of barrier or Lyapunov residual violation, separated by identified mode.
Pareto operational impact	False-safe upper bound, expected unserved energy, renewable curtailment, recovery time, trips, and equipment wear; compare whole fronts rather than one weighted cost.
Latency compliance	p50/p95/p99 and maximum response time for nominal and escalated paths, plus compute/memory utilization; verify (8).
Communication resilience	Detection delay, false acceptance, fallback correctness, and physical excursions under jitter, correlated loss, stale data, and hold-last one/two-cycle command loss; verify (14).
Irreducible pessimism	Count/rate/cost of all-system safe actions when all modes are low-confidence and calibration is out-of-support; require $\rho_{\text{TP}} \leq 5\%$.
Adaptation efficiency	Primary: false-safe learning curve against ΔN_{acq} with common N_{base}^0 . Secondary: full lifecycle ledger in (10), coverage, source-regime retention, and update/rollback counts.
Large-system scalability	Number of critical interfaces, DIM classes, calibration cells, concurrent escalations, and per-interface nominal/escalated p99/max latency and coverage.

decisive ablations are H1 without the calibrated residual-to-safe-set link, H1 without hold-last tolerance, H2 with fixed rather than gated constrained online weights, and the cascade replaced by the serial stack in (B6). These show respectively whether credibility is cosmetic, whether actuation loss invalidates a nominal certificate, whether online adaptation is safe under drift, and whether the method meets its claimed deadline without suppressing required physics checks.

C. Expected Outcomes and Interpretation

The expected outcome is not perfect prediction. A successful H1 system should move ambiguous cases from unsafe “false safe” decisions to either conservative but feasible actions or escalation, while meeting (8), satisfying hold-last tolerance in (14), keeping irreducible pessimism at or below the preregistered limit in (15), and improving the Pareto frontier in (13). A successful H2 system should concentrate expensive EMT labels near the intersection of novelty, error risk, certificate fragility, and operational consequence, while a deadband-gated constrained update maintains the rolling safety set. Its primary efficiency curve uses ΔN_{acq} against a common deployment baseline; its full lifecycle ledger remains visible through (10). If a method lowers false-safe events only through disproportionate shedding, misses its deadline, loses physical-residual coverage under OOD, invokes all-system safing too often, fails hidden communication stress, or cannot scale the critical-interface branch, the central hypotheses are

not supported. These negative results identify which bridge in (1) failed.

IX. READINESS, RISK, AND REPRODUCIBILITY

Research readiness should not be confused with a claim that the hypotheses are true. We distinguish: (i) *corpus readiness*, supplied by the 45-paper corpus and its coverage of all six SH functions; (ii) *evidence readiness*, supplied by direct theoretical, simulation, EMT pseudo-label, and HIL-falsification tests; (iii) *deployment readiness*, requiring a declared deadline, branchwise latency measurements, physical residual coverage, hold-last behavior under communication loss, an irreducible-pessimism audit, and a declared critical-interface scope; and (iv) *scientific assessment*, which remains “inconclusive” until the planned comparisons are completed. Reproducibility requires versioned network models, fault/control/communication parameter grids, simulator versions, train/validation/test scenario splits, random seeds, calibration-cell definitions and counts, common N_{base}^0 and all incremental-label ledger entries, certificate residual definitions, frozen HIL cases, meta-update gate/acceptance/rollback logs, Pareto-front generation code, critical-interface selection code, and all failed cases. Results should report both successful and failed certificate checks rather than silently excluding unstable, OOD, or delayed-data events.

X. CONCLUSION

The literature corpus supports a focused research programme beyond another high-accuracy TSA classifier. Its strongest unexploited opportunity is a deployable loop from calibrated physical residuals and credibility to a hybrid certificate, from that certificate to a latency-bounded action, and from failure risk back to high-value EMT labels. H1 maps credibility to physical margins and must improve a safety-disruption frontier without hiding behind excessive curtailment, an undelivered command, a missed deadline, or frequent all-system safing. H2 directs marginal post-deployment EMT investment to certificate-fragile states and must adapt to drift without weakening an independent safety hold-out or oscillating through noisy updates; its common calibration reserve remains visible as lifecycle cost rather than a selector penalty. Their value is conditional and falsifiable through physical coverage, false-safe rate, latency, certificate validity, hold-last tolerance, irreducible-pessimism rate, Pareto comparison, calibration-label accounting, critical-interface scalability, and hidden communication-stress testing across genuinely unseen operating regimes.

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